D 50663 **(Pages : 4)** **Name.....**

Reg. No.....

FIFTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2023

Mathematics

MTS 5D 03—LINEAR MATHEMATICAL MODELS

(2019 Admissions only)

Time : Two Hours Maximum : 60 Marks

Section A

Answer any number of questions.
Each question carries 2 *marks.*
Ceiling is 20.

1. Find slope of the line $y - 3x - 2 = 0$.

2. Write the equation of the line passing through $(2, -5)$ parallel to $2x - y = -4$.

3. Solve the system of equations $\frac{x}{2} + y = \frac{3}{2}, \frac{x}{3} + y = \frac{1}{3}$ using echelon method.

4. Replace R_2 by $\frac{1}{5}R_2$ and R_3 by $\frac{1}{2}R_3$ in the following augmented matrix : $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 30 \\ 0 & 5 & 0 & 10 \\ 0 & 0 & 6 & 162 \end{array}\right]$.

5. Compute $\begin{bmatrix} 8 & 5 & 2x \\ 1 & 3 & 5 \end{bmatrix} - \begin{bmatrix} 2 & 1 & x \\ 3 & 1 & 1 \end{bmatrix}$.

6. Let $A = \begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix}, B = \begin{bmatrix} 5 & 1 \\ 0 & 1 \end{bmatrix}$. Find $2A + 3B$.

7. Graph the linear inequality $y \geq x - 3$.

8. Give an example of a minimization LPP with 2 variables and 4 constraints.

Turn over

9. Convert the inequality constraint $x_1 + 2x_2 \leq 6$ into equality by adding a slack variable.

10. Sketch the feasible region for the maximization problem :

Maximize $Z = 3x - 2y$ subject to $x \geq 0, y \geq 0, y \leq 5$.

11. Find the transpose of $A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 2 \\ 3 & 5 & 9 \end{bmatrix}$.

12. Write the minimization problem corresponding to the augmented matrix : $\left[\begin{array}{cc|c} 1 & 3 & 6 \\ 2 & 1 & 3 \\ \hline 3 & 2 & 0 \end{array}\right]$.

Section B

Answer any number of questions.
Each question carries 5 marks.
Ceiling is 30.

13. The percentage of adults in the United States, without health insurance increased at a roughly linear rate from 1999, when it was 17.2 %, to 2008, when it was 20.3 %. Determine a linear equation that approximates the percentage of adults in the United States without health insurance in terms of time t, where t represents the number of years since 1990. If this trend were to continue, in what year would the percentage of adults without health insurance be at least 25 % ?

14. A flight leaves New York at 8 P.M. and arrives in Paris at 9 A.M. (Paris time). This 13-hour difference includes the flight time plus the change in time zones. The return flight leaves Paris at 1 P.M. and arrives in New York at 3 P.M. (New York time). This 2-hour difference includes the flight time minus time zones, plus an extra hour due to the fact that flying westward is against the wind. Find the actual flight time eastward and the difference in time zones.

15. Use Gauss-Jordan method to solve the system of equations :

$$x - z = -3$$
$$y + z = 9$$
$$-2x + 3y + 5z = 33.$$

16. Find A^{-1} if $A = \begin{bmatrix} 4 & -6 & -3 \\ -1 & 2 & 1 \\ -2 & 3 & 2 \end{bmatrix}$ using row operations.

17. Graph the feasible region for the following system :

$$2x + 3y \leq 12$$
$$2x + 3y > 3$$
$$3x + y < 4$$
$$x \geq 0$$
$$y \geq 0.$$

Tell whether the region is bounded or unbounded.

18. Use graphical methods to solve each linear programming problem :

Minimize $z = 3x + 2y$ subject to

$$10x + 7y \leq 42$$
$$4x + 10y \geq 35$$
$$x \geq 0$$
$$y \geq 0.$$

19. Using the circled element, pivot and read the solutions from the result :

$$\begin{array}{c} \begin{matrix} x_1 & x_2 & x_3 & s_1 & s_2 & s_3 & z \end{matrix} \\ \left[\begin{array}{ccccccc|c} 4 & 2 & 3 & 1 & 0 & 0 & 0 & 22 \\ 2 & 2 & \textcircled{5} & 0 & 1 & 0 & 0 & 28 \\ 1 & 3 & 2 & 0 & 0 & 1 & 0 & 45 \\ \hline -3 & -2 & -4 & 0 & 0 & 0 & 1 & 0 \end{array}\right] \end{array}.$$

Section C

Answer any **one** *question.*
The question carries 10 *marks.*

20. In a publishing company, the fixed cost to produce a typical book volume is \$525, and the total cost to produce 1000 copies of the book is \$2675. The books sell for a price of \$4.95 each.

    (a) Find the linear cost function for the book production.

    (b) How many such books must the company produce and sell in order to break even ?

    (c) How many books must the company produce and sell to make a profit of \$1000 ?

21. Use duality to solve the linear programming problem :

$$\begin{aligned} \text{Minimize } w &= 3y_1 + 2y_2 \\ \text{subject to } & 2y_1 + 3y_2 \geq 60 \\ & y_1 + 4y_2 \geq 40 \\ & y_1, y_2 \geq 0. \end{aligned}$$

(1 × 10 = 10 marks)

D 50663-A **(Pages : 4)** **Name.....**

Reg. No.....

FIFTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2023

Mathematics

MTS 5D 03—LINEAR MATHEMATICAL MODELS

(2019 Admission only)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes	**Total No. of Questions : 15**	**Maximum : 15 Marks**

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 15.
2. The candidate should check that the question paper supplied to him/her contains all the 15 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 5D 03—LINEAR MATHEMATICAL MODELS

(Multiple Choice Questions for SDE Candidates)

1. What is the equation of line passing through the points $(1, -3)$ and $(3, 1)$?

   (A) $y = 2x - 7$. (B) $y = -2x - 7$.

   (C) $y = -2x + 5$. (D) $y = 2x - 5$.

2. Equation of a line whose slope is $\frac{-3}{2}$ and x-intercept is 1 :

   (A) $3x + 2y = 3$. (B) $2x - 3y = 1$.

   (C) $2x + 3y = 3$. (D) $2x + 3y = -3$.

3. P : Two lines having different slopes is always intersect.

   Q : Two distinct lines having same slopes is never intersect.

   (A) P is true and Q is false. (B) P is false and Q is true.

   (C) P and Q are false. (D) P and Q are true.

4. Find the equation of the line that passes through the point $(3, 7)$ and is perpendicular to the line $5x - y = 4$:

   (A) $x + 5y = 38$. (B) $5x - y = 8$.

   (C) $x - 5y = -32$. (D) $x + 5y = 38$.

5. Find the equation of the line that passes through the point $(1, 0)$ and is perpendicular to the line $y = 0$:

   (A) $x = 0$. (B) $x = 1$.

   (C) $x = -1$. (D) None of the above.

6. $\begin{bmatrix} 5 & -6 \\ 8 & 9 \end{bmatrix} + \begin{bmatrix} -4 & 6 \\ 8 & -3 \end{bmatrix} =$ _____.

(A) $\begin{bmatrix} -1 & 0 \\ 16 & 6 \end{bmatrix}$.

(B) $\begin{bmatrix} 1 & 0 \\ 16 & 6 \end{bmatrix}$.

(C) $\begin{bmatrix} -1 & 12 \\ 16 & -6 \end{bmatrix}$.

(D) $\begin{bmatrix} -1 & 0 \\ 16 & -6 \end{bmatrix}$.

7. Let A be a matrix of order $m \times n$ and B be a matrix of order $p \times q$. Then A + B is defined if and only if

(A) $m = p$ and $n = q$.

(B) $m = p$ and $n \neq q$.

(C) $m \neq p$ and $n = q$.

(D) $m \neq p$ and $n \neq q$.

8. P : Matrix addition is commutative.

Q : Matrix addition is Associative.

(A) P is true and Q is false.

(B) P is false and Q is true.

(C) P and Q are false.

(D) P and Q are true.

9. P : Matrix multiplication is commutative.

Q : Matrix multiplication is Associative.

(A) P is true and Q is false.

(B) P is false and Q is true.

(C) P and Q are false.

(D) P and Q are true.

10. $A = \begin{bmatrix} 5 & -6 \\ 8 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 6 \\ 8 & -3 \end{bmatrix}$. Then AB = _____.

(A) $\begin{bmatrix} -68 & 48 \\ 40 & 19 \end{bmatrix}$.

(B) $\begin{bmatrix} -68 & 48 \\ -104 & 21 \end{bmatrix}$.

(C) $\begin{bmatrix} -68 & 42 \\ 40 & 21 \end{bmatrix}$.

(D) $\begin{bmatrix} -68 & 48 \\ 40 & 21 \end{bmatrix}$.

11. Let $AX = B$ be a system of linear equations where A is invertible square matrix, Then :

(A) $X = AB$. (B) $X = BA$.

(C) $X = A^{-1}B$. (D) $X = BA^{-1}$.

12. Let $AX = B$ be a system of linear equations where A is invertible square matrix, Then it has _____.

(A) Unique solution. (B) Two solutions.

(C) Infinitely many solutions. (D) No solution.

13. P : If an optimum value of the objective function exists, it will occur at one or more of the corner points of the feasible region.

Q : If an optimum value of the objective function exists, it will occur at one of the corner points of the feasible region.

(A) P is true and Q is false. (B) P is false and Q is true.

(C) P and Q are false. (D) P and Q are true.

14. If $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ then A^{-1} = _____.

(A) $\begin{bmatrix} \frac{-1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$. (B) $\begin{bmatrix} \frac{1}{2} & \frac{-1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$.

(C) $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{-1}{2} & \frac{1}{2} \end{bmatrix}$. (D) $\begin{bmatrix} \frac{-1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{-1}{2} \end{bmatrix}$.

15. If $A = \begin{bmatrix} 1 & -3 \\ -1 & 4 \end{bmatrix}$ then $-5A$ = _____.

(A) $\begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}$. (B) $\begin{bmatrix} 6 & 2 \\ 5 & 9 \end{bmatrix}$.

(C) $\begin{bmatrix} 5 & -15 \\ 0 & 20 \end{bmatrix}$. (D) $\begin{bmatrix} -5 & 15 \\ 5 & -20 \end{bmatrix}$.